Spark SQL

Introduction

- Spark SQL is a module for structured data processing, upon which the higher-level structured APIs are built.
- Lets us query data using JDBC/ODBC connectors from external business intelligence sources such as Tableau, Power BI, Snowflake, or from RDBMS such as MySQL and PostgreSQL.
- Offers an interactive shell to issue SQL queries on our structured data.
  - We can interact with the SQL interface via a programming language and the data set will be returned as a `Dataset`/`DataFrame`.
- When computing a result the same execution engine is used, independent of which API/language you are using to express the computation. This unification means that developers can easily switch back and forth.

Examples

- Example 1: Using SQL with Spark
- Example 2: Manipulating databases directly using SQL in Spark
- Example 3: Java and SQL in Spark